

Pearl Nibebadaar Sandoh KUURIDONG

Curriculum Vitae

Personal Details

Date of Birth: 18 June 2000

Address: Kigali, Rwanda

Phone: +233204952684, +250796884511

Nationality: Ghanaian

Email: pearl.kuuridong@aims.ac.rw

Website: Pearl Kuuridong

Education

2025 – 2026

MSc in Mathematical Sciences

African Institute for Mathematical Sciences (AIMS), Kigali, Rwanda

Thesis Title:

Combinatorial Optimization with Graph Neural Networks (Completed).

Thesis Summary:

Tested the GCON self-supervised GNN framework on Maximum Independent Set and Balanced Graph Partitioning, deriving problem-specific loss functions and decoders, and identifying a ranking reversal in the MIS loss landscape and a balance-constraint failure in spectral baselines for graph partitioning.

Relevant Courses:

Graphs, Matroids, and Designs, Data Driven Optimization, Algorithms and Data Structures, Statistical Machine Learning, Introduction to Neural Networks, Numerical Linear Algebra, Mathematical Problem Solving.

2018 – 2022

BSc in Mathematics and Computer Science

University of Ghana.

Thesis Title:

Applying Persistent Homology to Topological Data Analysis.

Thesis Summary:

Studied the algebraic and topological foundations of persistent homology, working through the underlying machinery connecting algebraic topology to multi-scale data analysis.

Relevant Courses:

Design and Analysis of Algorithms, Data Structures and Algorithms, Programming with C++, Abstract Algebra, Linear Algebra, Probability Theory.

Skills

Languages: English (Fluent), Dagaare (Mother tongue)

Software: $\text{\LaTeX}$, R, SQL, Python, Bash, PyTorch.

Experience

2022 – 2023

Teaching Assistant: Abstract Algebra, General Topology, Linear Algebra
University of Ghana

- Graded coursework for Linear Algebra
- Supported students in Abstract Algebra and General Topology through informal availability for questions outside of scheduled hours

2023 – 2025

Data Engineer
Cape Wesley Consults

- Wrote SQL to extract and structure data from a bank's database for an in-progress automated compliance reporting system for the Bank of Ghana
- Used Apache Solr to feed structured data to the frontend of a reporting platform for a regulatory authority client
- Wrote Bash scripts to automate monitoring tasks on the same project, including checks for new database entries and triggering notifications
- Worked primarily in SQL, Python, and Bash across client engagements

Awards

- 2025 – Mastercard Foundation Scholar at AIMS Rwanda
- 2026 – AIMS Rwanda Exceptional Excellence Award (Best Female Display of the AIMS Value of Excellence)

References

Dr. Olakunle Abawonse

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Professor Michael Perlmutter

Assistant Professor, Department of Mathematics, Boise State University, Boise, ID, USA
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Dr. Donald Woukeng Feudjio

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Prof. Blaise Tchapnda

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Dr. Kenneth Dadedzi

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